# Sergei Volkov

## Senior Data & MLOps Engineer

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**Work Authorization:** EU Blue Card holder (>2 yrs) · Unrestricted German Work Permit (incl. Freiberufler / Commercial)

**Senior Data & MLOps Engineer** and active NixOS package maintainer with 6+ years designing and scaling production data platforms across gaming analytics (Playrix, 50M+ DAU), enterprise e-commerce (Zalando), and ML experimentation (AppIQ). Specialized in self-hosted lakehouses (DuckDB/DuckLake on AWS S3, Delta Lake), orchestration (Apache Airflow, CI/CD), and low-latency ML model serving. Official package maintainer in `nixpkgs` (Apache Airflow, Julia, Databricks CLI). Co-author of 5 peer-reviewed publications in computational fluid dynamics.

## WORK EXPERIENCE

### Freelance Data / MLOps Engineer | AppIQ
Jan 2026 – Present

*Architecture & Implementation of ML-driven Experimentation Platform*

- Architected and deployed a self-hosted Lakehouse (**DuckDB/DuckLake on AWS S3**) via **Terraform**, delivering sub-second analytical query latency while eliminating Spark operational overhead.
- Built low-latency ML model serving infrastructure and automated training/evaluation pipelines for real-time model inference.
- Designed high-throughput event ingestion and user tracking pipelines, enabling automated A/B test assignment and real-time metric calculation.
- Standardized declarative infrastructure-as-code and reproducible deployment workflows using **Terraform** and **Nix**.

### Sabbatical & Open Source Engineering | Self-Directed
Apr 2025 – Sep 2025

*Open Source Contributions & Advanced Technical Upskilling*

- Dedicated sabbatical for intensive open-source contributions to **NixOS** (`nixpkgs`) and packages' upstream, packaging and maintaining **Apache Airflow**, **Julia**, and **Databricks CLI**.
- Conducted architectural research on embedded analytical engines (DuckDB) and prepared for advanced machine learning training.

### Data Engineer | Zalando SE
Oct 2024 – Mar 2025

*Central Data Platform & Analytics Engineering Services*

- Developed and maintained business-critical ETL/ELT pipelines using **Managed Workflows for Apache Airflow (MWAA)** and **Databricks Delta Lake**, supporting 15+ internal stakeholder teams.
- Engineered automated CI/CD deployment pipelines for internal Python helper packages and notebook workloads across staging and production Databricks environments.
- Conducted fraudulent customers detection dataset design.
- Provided code reviews, data QA, and architectural support for cross-functional data consumers.

### Data Scientist & Cloud Architect | Crocus Labs GmbH
Aug 2023 – Feb 2024

*IoT Platform Architecture & Computational Photometry*

- Architected scalable cloud infrastructure on **AWS (IoT Core, Lambda, S3, DynamoDB)** for real-time telemetry ingestion from smart greenhouse LED hardware.
- Developed a Python raytracing simulation to model 3D light distribution and optimize LED chip array layouts based on raw photometric emitter data.
- Engineered a containerized IoT test harness with virtualized sensor clients and an **MQTT broker**, enabling end-to-end integration testing without physical hardware.

**Data Quality Engineer / Analytics Engineer** | **Playrix** Nov 2019 – Feb 2023
*Promoted across 3 roles (Data Analyst -> QA Engineer -> Data Quality Engineer) for 50M+ DAU platforms*

- Led data quality assurance during the migration of legacy Big Data pipelines to a modern lakehouse architecture across **Spark, Hive, Redshift, and Snowflake**.
- Built dynamic SQL metaprogramming validation frameworks that automated data anomaly detection across billions of daily in-game event records.
- Designed and optimized executive **Tableau** dashboards and underlying dimensional data marts tracking player retention, monetization, and A/B test experiments.
- Managed a QA sub-team, establishing automated regression test suites and data verification protocols for engineering releases.

**Research Engineer** | **Leibniz Institute of Freshwater Ecology (IGB Berlin)** Jun 2021 – Mar 2023
*Computer Vision & Environmental Data Infrastructure*

- Architected containerized computational environments (**Docker, CMake, MATLAB, OpenCV**) to modernize and execute a legacy 3D particle-tracking codebase.
- Developed computer vision data processing pipelines in **Python** to extract 3D velocity vectors from underwater stereo-video recordings of particulate biomass.
- Administered remote Linux compute nodes and automated data extraction workflows for gigabyte-scale video datasets.

**Research Data Analyst** | **Northern Water Problems Institute (NWPI)** Jan 2016 – Nov 2024
*Telemetry Processing & Numerical Modeling*

- Co-authored 5 peer-reviewed papers on convective turbulence and thermal stratification; processed long-term ADCP and thermistor telemetry.
- Built numerical analysis and data processing frameworks in **Python** and **MATLAB** for long-term limnological time-series data.

## TECHNICAL COMPETENCIES

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|---|---|
| **Data & Lakehouse:** | Spark (PySpark), Databricks, DuckDB / DuckLake, Snowflake, BigQuery, dbt, Delta Lake, Redshift |
| **Orchestration & Infra:** | Apache Airflow (**NixOS Maintainer**), Docker, Podman, Kubernetes, Terraform, Nix/NixOS, CI/CD |
| **Cloud & Storage:** | AWS (S3, Lambda, IoT Core, MWAA, DynamoDB), GCP, PostgreSQL, MySQL |
| **ML & Analytics:** | Scikit-Learn, PyTorch, TensorFlow, MLflow, Model Serving, A/B Testing, Time-Series Analysis |
| **Languages:** | Python (NumPy, Pandas, SciPy), SQL, Bash, Julia, MATLAB |

## EDUCATION, QUALIFICATIONS & COMMUNITY

**Data Science & Machine Learning Bootcamp** Oct 2025 – Dec 2025
Ironhack

**Specialist Degree in Energy Supplies (M.Sc. Equivalent)** 2010 – 2015
Petrozavodsk State University

**Machine Learning Specialization** 2017
Stanford University on Coursera

**Open Source:** Official Package Maintainer for NixOS (`nixpkgs`: Apache Airflow, Julia, Databricks CLI, plus 6 Python libraries, system utilities) · OpenStreetMap Contributor
**Languages:** English (C1/C2 Full Professional), German (B2 Professional Working), Russian (Native)
*Complete bibliography of 5 peer-reviewed publications and academic projects available in attached addendum or upon request.*